Lecture 4. Discrete Random Variables

Expectation, variance, binomial, Poisson, geometric, negative binomial, hypergeometric

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4.1 Random variables and distribution functions

A **random variable** is a real function on the sample space. Its **(cumulative) distribution function** is $F(x) = P(X \le x)$. A random variable is **discrete** if it takes at most countably many values $x_1, x_2, \ldots$; its **probability mass function** is $p(x) = P(X = x)$, with $\sum_i p(x_i) = 1$ and $F(a) = \sum_{x_i \le a} p(x_i)$, a step function.

Proposition 4.1 (Properties of F)

F is nondecreasing, right-continuous, $F(-\infty) = 0$ and $F(\infty) = 1$. Moreover $P(a < X \le b) = F(b) - F(a)$ and $P(X < b) = \lim_{x \uparrow b} F(x)$.

Proof. Monotonicity is Proposition 2.1(b). If $x_n \downarrow b$, the events $\{X \le x_n\}$ decrease to $\{X \le b\}$; if $x_n \uparrow \infty$ they increase to S; if $x_n \uparrow b$ the events $\{X \le x_n\}$ increase to $\{X < b\}$. Apply continuity (Proposition 2.3). ■

Example 4.1 (Coupon collecting)

Each purchase gives one of N coupon types, equally likely and independently. Let T be the number of purchases needed to collect all types. With A_j the event that type j is missing after n purchases, inclusion–exclusion gives

$$P(T > n) = P\Big(\bigcup_j A_j\Big) = \sum_{i=1}^{N-1} (-1)^{i+1} \binom{N}{i} \Big(\frac{N-i}{N}\Big)^n,$$

and $P(T = n) = P(T > n-1) - P(T > n)$.

4.2 Expectation

Motivation. A random variable's long-run average is its expectation. It summarizes the distribution in one number and is the basis of fair prices, variance, and the laws of large numbers.

Definition (Expected value)

$E[X] = \sum_x x p(x)$, provided the sum converges absolutely.

It is the long-run average of repeated observations and the centre of gravity of the mass function. For the indicator I_A of an event, $E[I_A] = P(A)$.

Proposition 4.2

(a) $E[g(X)] = \sum_x g(x)p(x)$.
(b) $E[aX + b] = aE[X] + b$.
(c) On a countable sample space, $E[X] = \sum_{s \in S} X(s)P(\{s\})$, and hence $E[\sum_i X_i] = \sum_i E[X_i]$.

Proof. (a) Group the values x by $y = g(x)$: $\sum_y y P(g(X) = y) = \sum_y \sum_{x: g(x)=y} g(x)p(x) = \sum_x g(x)p(x)$.

(b) is a special case.

(c) Group outcomes s by the value $X(s)$; then linearity of the sum over s gives the second claim. ■

In general $E[g(X)] \ne g(E[X])$.

Example 4.2 (Which question first?)

A quiz contestant may answer question 1 or 2 first, and goes on to the other only if the first is answered correctly. Question i is answered correctly with probability p_i, independently, and is worth V_i. Answering 1 first gives expected winnings $p_1(1-p_2)V_1 + p_1p_2(V_1+V_2) = p_1V_1 + p_1p_2V_2$; the other order gives $p_2V_2 + p_1p_2V_1$. So question 1 should come first iff

$$\frac{p_1V_1}{1-p_1} > \frac{p_2V_2}{1-p_2}.$$

Example 4.3 (The friendship paradox)

In a network, let d_i be the number of friends of person i, and pick a random friendship (edge) and then one of its two ends at random. That person has D^* friends, and $P(D^* = k) = k\,P(D=k)/E[D]$, where D is the degree of a uniformly chosen person: a person with k friends is k times as likely to be reached. Hence

$$E[D^*] = \frac{E[D^2]}{E[D]} \geq E[D],$$

with equality only if all degrees are equal (by the variance inequality below). "Your friends have more friends than you", on average.

4.3 Variance

Definition (Variance)

With $\mu = E[X]$, $\operatorname{Var}(X) = E[(X-\mu)^2] = E[X^2] - \mu^2$. The standard deviation is $\sqrt{\operatorname{Var}(X)}$.

The second formula follows by expanding the square and using Proposition 4.2. As $\operatorname{Var}(X) \geq 0$, we get $E[X^2] \geq (E[X])^2$. Also $\operatorname{Var}(aX+b) = a^2 \operatorname{Var}(X)$. For a fair die, $E[X] = 7/2$, $E[X^2] = 91/6$, so $\operatorname{Var}(X) = 35/12$.

4.4 Bernoulli and binomial random variables

$X \sim \operatorname{Bin}(n,p)$ counts successes in n independent trials with success probability p:

$$p(k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, \ldots, n.$$

Each specific sequence with k successes has probability $p^k(1-p)^{n-k}$ and there are $\binom{n}{k}$ such sequences. $\operatorname{Bin}(1,p)$ is the Bernoulli distribution.

Proposition 4.3

If $X \sim \operatorname{Bin}(n,p)$ then $E[X^k] = np\,E[(Y+1)^{k-1}]$ with $Y \sim \operatorname{Bin}(n-1,p)$. Hence $E[X] = np$ and $\operatorname{Var}(X) = np(1-p)$.

Proof. Using $i\binom{n}{i} = n\binom{n-1}{i-1}$,

$$E[X^k] = \sum_{i=1}^{n} i^{k-1}\, n \binom{n-1}{i-1} p^i (1-p)^{n-i} = np \sum_{j=0}^{n-1} (j+1)^{k-1} \binom{n-1}{j} p^j (1-p)^{n-1-j}.$$

$k=1$ gives np; $k=2$ gives $E[X^2]=np\{(n-1)p+1\}$, so $\mathrm{Var}(X)=np(1-p)$. ■

Since $p(k)/p(k-1)=\frac{(n-k+1)p}{k(1-p)}$, the pmf increases while $k<(n+1)p$ and then decreases: it is unimodal with mode $\lfloor (n+1)p \rfloor$. The ratio also gives a stable recursion for computing the cdf.

Example 4.4 (Chuck-a-luck)

A player bets 1 on a number; three dice are rolled and the player wins i if the number appears $i \geq 1$ times, and loses 1 otherwise. The number of appearances is $\mathrm{Bin}(3, 1/6)$, so the expected gain is

$$-\tfrac{125}{216}+1\cdot\tfrac{75}{216}+2\cdot\tfrac{15}{216}+3\cdot\tfrac{1}{216}=-\tfrac{17}{216}\approx -0.079.$$

Example 4.5 (k-out-of-n systems)

A system of n components, each working independently with probability p, works if at least half of them work. For which p is a 5-component system better than a 3-component one? $P_5=\binom{5}{3}p^3q^2+5p^4q+p^5$ and $P_3=3p^2q+p^3$. One finds $P_5-P_3=3p^2(p-1)^2(2p-1)$, so the larger system is better exactly when $p>\frac{1}{2}$.

4.5 The Poisson random variable

Motivation. Counts of rare events (typos, arrivals, defects) are approximately Poisson: a binomial with large n and small p converges to it. Its single parameter is both mean and variance.

$X\sim\mathrm{Poisson}(\lambda)$ if $p(k)=e^{-\lambda}\lambda^k/k!$, $k=0,1,\ldots$; then $E[X]=\mathrm{Var}(X)=\lambda$ (by the same shift trick: $E[X]=\lambda\sum_j e^{-\lambda}\lambda^j/j!$ and $E[X^2]=\lambda E[X+1]$).

Proposition 4.4 (Poisson approximation)

If $n\to\infty$ and $p\to 0$ with $np=\lambda$ fixed, then $\binom{n}{k}p^k(1-p)^{n-k}\to e^{-\lambda}\lambda^k/k!$.

Proof. $\binom{n}{k}p^k(1-p)^{n-k}=\dfrac{n(n-1)\cdots(n-k+1)}{n^k}\cdot\dfrac{\lambda^k}{k!}\cdot\dfrac{(1-\lambda/n)^n}{(1-\lambda/n)^k}$, and the three factors tend to 1, $\lambda^k/k!$ and $e^{-\lambda}$. ■

So the number of rare events among many trials (misprints on a page, people over 100, wrong numbers dialled) is approximately Poisson. The approximation remains good when the trials are only *weakly dependent* and success probabilities are small but unequal, with $\lambda=\sum p_i$ (the Poisson paradigm).

Example 4.6 (Poisson paradigm)

(a) Matching: the events "person i gets own hat" have probability $1/N$ and are weakly dependent, so the number of matches is approximately Poisson(1), as found exactly in Example 2.4.
(b) Birthdays: each of the $\binom{n}{2}$ pairs shares a birthday with probability $1/365$, and pair events are weakly dependent, so $P(\text{no shared birthday})\approx\exp\{-n(n-1)/730\}$. For $n=23$ this is $e^{-253/365}\approx 0.500$; the exact value is 0.493.

Example 4.7 (Poisson process)

Events (earthquakes, arrivals) occur in time so that (i) counts in disjoint intervals are independent, (ii) in a short interval of length h, one event occurs with probability $\lambda h + o(h)$ and two or more with probability $o(h)$. Splitting $[0, t]$ into n pieces, the count $N(t)$ is approximately $\mathrm{Bin}(n, \lambda t/n)$, and letting $n \to \infty$,

$$P(N(t) = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}.$$

4.6 Other discrete distributions

Geometric. The number X of trials up to and including the first success: $p(k) = (1-p)^{k-1}p$, $k \geq 1$, and $P(X > k) = (1-p)^k$. With $q = 1 - p$, conditioning on the first trial gives $E[X] = p \cdot 1 + q(1 + E[X])$, so $E[X] = 1/p$; similarly $E[X^2] = p + q\,E[(1+X)^2]$ gives $E[X^2] = (2-p)/p^2$ and $\mathrm{Var}(X) = q/p^2$.

Negative binomial. The number X of trials needed for r successes: the r-th success at trial k requires $r-1$ successes in the first $k-1$ trials, so

$$p(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k \geq r, \qquad E[X] = \frac{r}{p}, \quad \mathrm{Var}(X) = \frac{r(1-p)}{p^2}.$$

(X is a sum of r independent geometric waiting times, so Lecture 7 gives the moments at once.)

Example 4.8 (Banach match problem)

A smoker carries two boxes of N matches and each time picks a box at random. When a box is first found empty, what is the probability that the other contains exactly k matches? Call choosing the left box a "success". The left box is found empty with k matches on the right when success number $N+1$ occurs at trial $N + 1 + N - k$. So the probability is $2\binom{2N-k}{N}(\frac{1}{2})^{2N-k+1} = \binom{2N-k}{N}(\frac{1}{2})^{2N-k}$, the factor 2 accounting for the symmetric case.

Hypergeometric. Draw n balls without replacement from N, of which m are white; the number X of white balls drawn has

$$P(X = i) = \frac{\binom{m}{i}\binom{N-m}{n-i}}{\binom{N}{n}}, \qquad E[X] = \frac{nm}{N}, \qquad \mathrm{Var}(X) = n\frac{m}{N}\Big(1 - \frac{m}{N}\Big)\frac{N-n}{N-1}.$$

With $p = m/N$ this is the binomial mean, and the variance carries the finite-population correction $(N-n)/(N-1)$. When N is large relative to n, X is approximately $\mathrm{Bin}(n, m/N)$.

Example 4.9 (Capture–recapture)

To estimate an unknown population size N, catch and tag m animals, release them, and later catch n; suppose i are tagged. The probability of this, $P_i(N) = \binom{m}{i}\binom{N-m}{n-i}/\binom{N}{n}$, satisfies

$$\frac{P_i(N)}{P_i(N-1)} = \frac{(N-m)(N-n)}{N(N-m-n+i)} \geq 1 \iff N \leq \frac{mn}{i},$$

so the value of N that makes the observation most likely (the maximum likelihood estimate) is $\lfloor mn/i \rfloor$: the tagged fraction in the sample, i/n, is set equal to m/N.

Zeta (Zipf). $p(k) = C/k^{\alpha+1}$, $k \geq 1$, with $C = 1/\zeta(\alpha+1)$; it models heavy-tailed frequencies such as word counts.

4.7 Expected value of sums

By Proposition 4.2(c), $E[X_1+\cdots+X_n] = E[X_1]+\cdots+E[X_n]$ for *any* random variables, independent or not. Writing a count as a sum of indicators is the main trick of the course.

Example 4.10

(a) The sum of n fair dice has mean $7n/2$.
(b) If trial i succeeds with probability p_i, the expected number of successes is $\sum p_i$.
(c) The hypergeometric mean: with I_j the indicator that white ball j is drawn, $E[X] = \sum_{j=1}^{m} P(I_j = 1) = m \cdot n/N$ (Example 2.1(b)).

Example 4.11 (Variance of the number of successes)

With $X = \sum_i I_i$, $X^2 = \sum_i I_i + \sum_{i \neq j} I_i I_j$, so $E[X^2] = \sum_i p_i + \sum_{i \neq j} P(I_i = I_j = 1)$. For independent trials with common p, $E[X^2] = np + n(n-1)p^2$ and $\operatorname{Var}(X) = np(1-p)$. For the hypergeometric, $P(I_i = I_j = 1) = \frac{m(m-1)}{N(N-1)}$ for two specific draws $i \neq j$, which gives the variance stated above.

4.8 Selected exercises

Motivation. These are Ross's theoretical exercises for Chapter 4 (book numbering). They practise the two basic computational moves for discrete distributions, re-indexing a binomial sum and conditioning on an underlying count, and connect the Poisson, gamma, binomial and negative binomial distributions.

Exercise 4.10 (A reciprocal moment of the binomial)

Let $X \sim \operatorname{Bin}(n, p)$ with $0 < p \leq 1$. Show that

$$E\Big[\frac{1}{X+1}\Big] = \frac{1-(1-p)^{n+1}}{(n+1)p}.$$

Solution.

Step 1. For $0 \leq i \leq n$,

$$\frac{1}{i+1}\binom{n}{i} = \frac{n!}{(i+1)!\,(n-i)!} = \frac{1}{n+1}\binom{n+1}{i+1}.$$

Step 2. With $q = 1-p$ and Proposition 4.2(a),

$$\begin{aligned} E\Big[\frac{1}{X+1}\Big] &= \sum_{i=0}^{n} \frac{1}{i+1}\binom{n}{i} p^i q^{n-i} = \frac{1}{(n+1)p} \sum_{i=0}^{n} \binom{n+1}{i+1} p^{i+1} q^{n-i} \\ &= \frac{1}{(n+1)p} \sum_{j=1}^{n+1} \binom{n+1}{j} p^j q^{n+1-j} = \frac{(p+q)^{n+1} - q^{n+1}}{(n+1)p}, \end{aligned}$$

by the binomial theorem (Theorem 1.3) with the $j=0$ term removed. Since $p+q=1$ this is the claim. ∎

Exercise 4.25 (Thinning a Poisson count)

The number N of events in a fixed period is Poisson(λ), and each event is counted with probability p, independently of the other events and of N. Show that the number M of counted events is Poisson(λp). If $\lambda = 10$ uranium deposits are expected in an area and each is discovered with probability $1/50$, find the probability that at most one deposit is discovered.

Solution.

Step 1. Given $N=n$, M is the number of successes in n independent trials with success probability p, so $P(M=k \mid N=n) = \binom{n}{k} p^k q^{n-k}$ for $k \le n$ and 0 for $k>n$, where $q=1-p$.

Step 2. By the law of total probability (Theorem 3.2, extended to the countable partition $\{N=n\}$ by countable additivity),

$$P(M=k) = \sum_{n=k}^{\infty} e^{-\lambda}\frac{\lambda^n}{n!}\cdot\frac{n!}{k!\,(n-k)!}p^k q^{n-k}$$
$$= e^{-\lambda}\frac{(\lambda p)^k}{k!}\sum_{j=0}^{\infty}\frac{(\lambda q)^j}{j!} = e^{-\lambda}\frac{(\lambda p)^k}{k!}e^{\lambda q} = e^{-\lambda p}\frac{(\lambda p)^k}{k!}.$$

Step 3. Intuitively, N is approximately $\mathrm{Bin}(n,\lambda/n)$ for large n, and then M is approximately $\mathrm{Bin}(n,\lambda p/n)$, whose Poisson limit is Poisson(λp) by Proposition 4.4.

Step 4. Here $\lambda p = 0.2$, so $P(M\le 1) = e^{-0.2}(1+0.2) \approx 0.982$. ∎

Exercise 4.26 (Poisson tails as gamma integrals)

Prove that for $\lambda>0$ and $n=0,1,2,\dots$,

$$\sum_{i=0}^{n} e^{-\lambda}\frac{\lambda^i}{i!} = \frac{1}{n!}\int_{\lambda}^{\infty} e^{-x}x^n\,dx.$$

Solution.

Step 1. Put $I_n = \frac{1}{n!}\int_\lambda^\infty e^{-x}x^n\,dx$, which is finite. Then $I_0 = \int_\lambda^\infty e^{-x}\,dx = e^{-\lambda}$, which is the claim for $n=0$.

Step 2. For $n\ge 1$ integrate by parts with $u=x^n/n!$ and $dv=e^{-x}dx$. Since $x^n e^{-x}\to 0$ as $x\to\infty$,

$$I_n = \Big[-e^{-x}\frac{x^n}{n!}\Big]_\lambda^\infty + \int_\lambda^\infty e^{-x}\frac{x^{n-1}}{(n-1)!}\,dx = e^{-\lambda}\frac{\lambda^n}{n!} + I_{n-1}.$$

Step 3. By induction on n, $I_n = \sum_{i=0}^{n} e^{-\lambda}\lambda^i/i!$. In words, if $X\sim$ Poisson(λ) and $Y\sim$ Gamma($n+1,1$), then $P(X\le n) = P(Y>\lambda)$: in a rate-one Poisson process, at most n events occur by time λ iff the $(n+1)$-st event occurs after λ. ∎

Exercise 4.28 (Negative binomial versus binomial)

Let X be negative binomial with parameters (r,p) and $Y\sim\mathrm{Bin}(n,p)$. Show that $P(X>n) = P(Y<r)$.

Solution.

Step 1. Consider one infinite sequence of independent trials with success probability p. Let X be the trial on which the r-th success occurs ($X = \infty$ if there is none) and Y the number of successes among the first n trials. Then X has the negative binomial distribution with parameters (r, p) (Section 4.6) and $Y \sim \mathrm{Bin}(n, p)$, so it suffices to show $\{X > n\} = \{Y < r\}$ as events.

Step 2. If $Y \geq r$, the first n trials contain at least r successes, so the r-th success occurs at or before trial n and $X \leq n$. Conversely, if $X \leq n$, the first $X \leq n$ trials already contain r successes, so $Y \geq r$. Hence $\{X \leq n\} = \{Y \geq r\}$, and taking complements gives the claim. ■